The Value of Storage in Electricity Distribution: The Role of Markets

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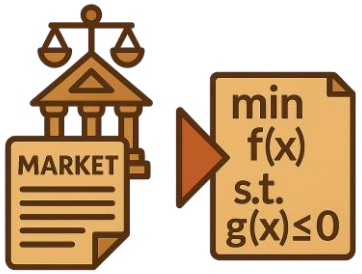
15 October 2025



Storage is deployed to meet peak electricity demand,

DIVE BRIEF

There once was a 48 MWh Tesla battery on Nantucket, which saved National Grid $120M in its budget

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Iulia Gheorghiu
Editor

National Grid

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MENU

10,000 BATTERIES

The BESS Is Up and Running on Race Point Road

But Eversource's Battery Energy Storage System never saw duty this winter

BY AMELIA ROTH-DISHY · MAY 17, 2023

PROVINCETOWN — For the first time in years, the Outer Cape managed to avoid major power outages last winter. There were storms — including on Dec. 23, when several East End streets were flooded with salt water, and on Feb. 4, when frozen pipes burst all over town — but they were not exacerbated by a loss of power that would have disabled sump pumps and heating systems as well.

but sits idle most of the time in deregulated markets

because of concerns over market distortion.

- Basis of deregulated markets: competition in electricity generation (Joskow and Schmalensee, 1983).

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- State legislation requires that generation be split from transmission and distribution, e.g., MA HB5117 (1998), in the hope to remedy a situation in which

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> Massachusetts General Court, 1997

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- The rationale was to increase competition in generation while avoiding any

> "**cross-subsidization** of competitive businesses from regulated businesses and **discriminatory policies** affecting access to distribution and transmission networks upon which all competitive suppliers depend."
> Joskow, 2008, p. 12

because of concerns over market distortion.

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Question to assess market distortions: How much battery storage capacity do utilities own?

EIA860, June 2025: US electricity generation capacity – 1300GW, batteries – 33GW, utility-owned – 3.4GW.

Some states experiment with allowing market participation to reduce retail prices.

STATE OF NEW YORK PUBLIC SERVICE COMMISSION

ORDER APPROVING UTILITY-OWNED ASSET PARTICIPATION IN WHOLESALE MARKETS

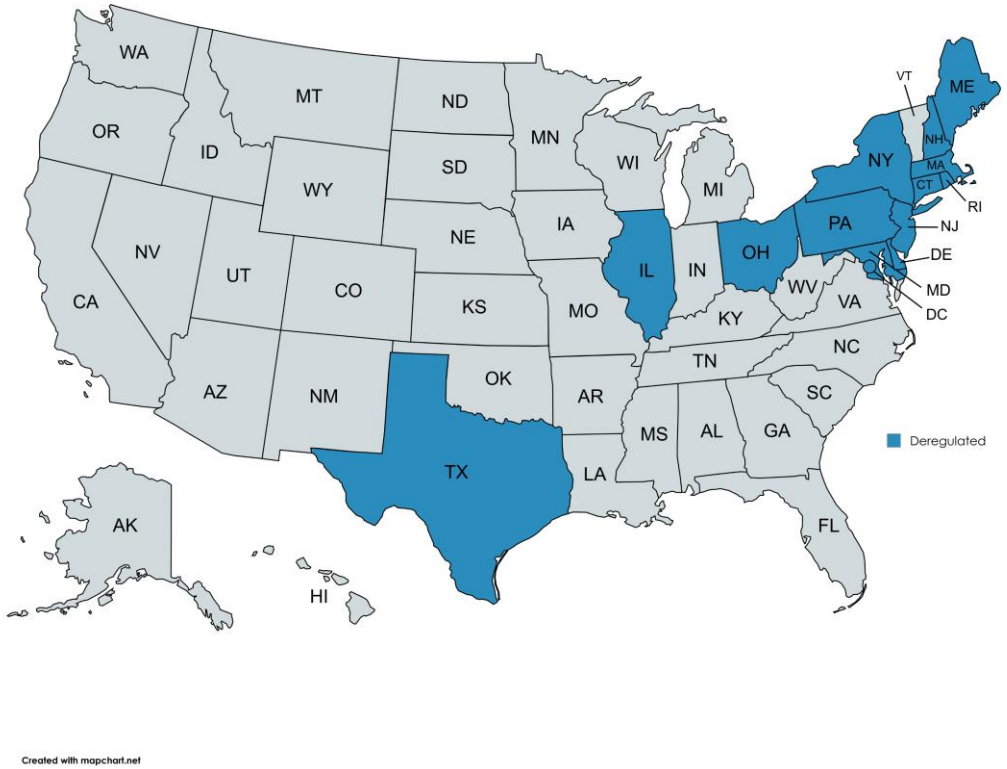
(Issued and Effective September 9, 2021)

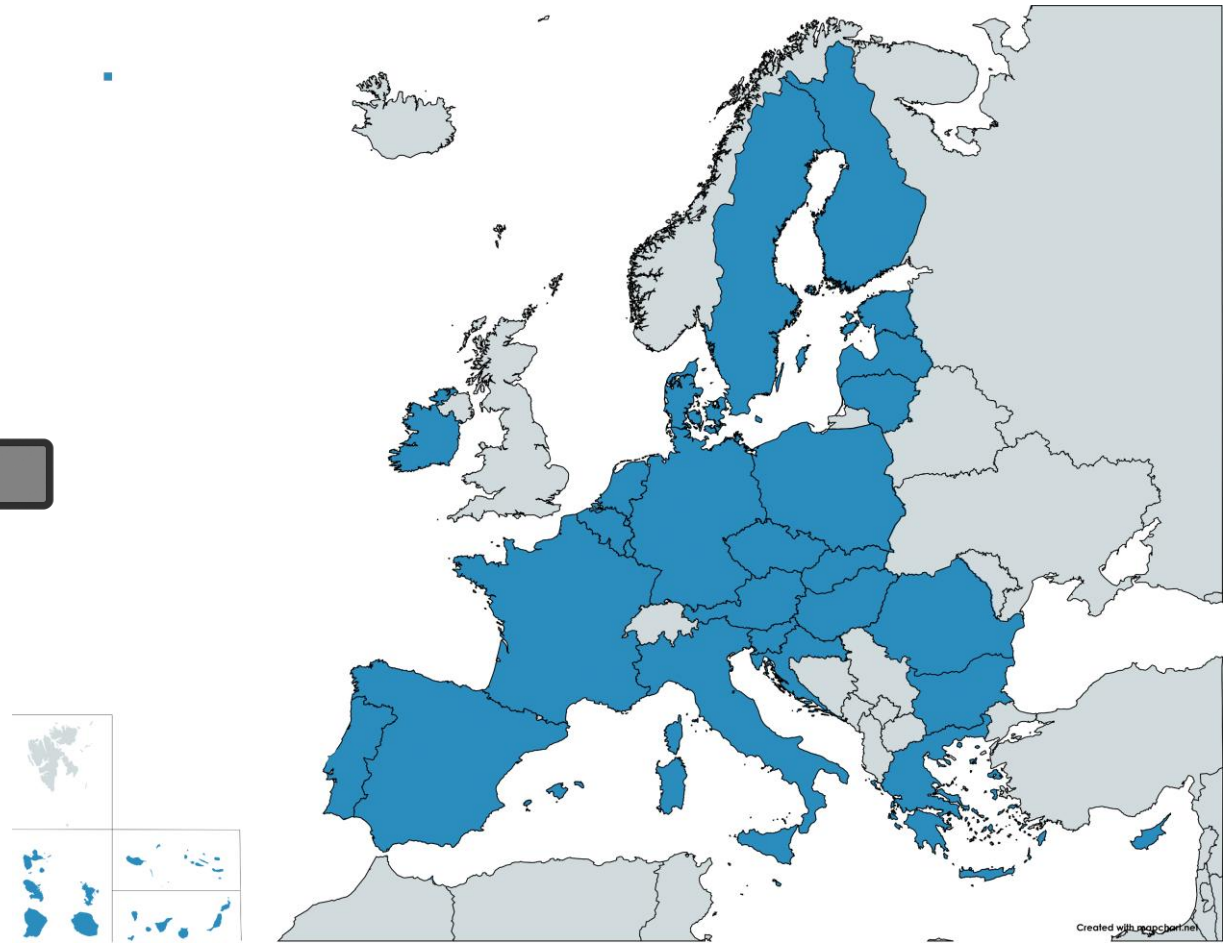
Pomona Substation, Orange & Rockland, NY, 3MW/12MWh

In 2024:

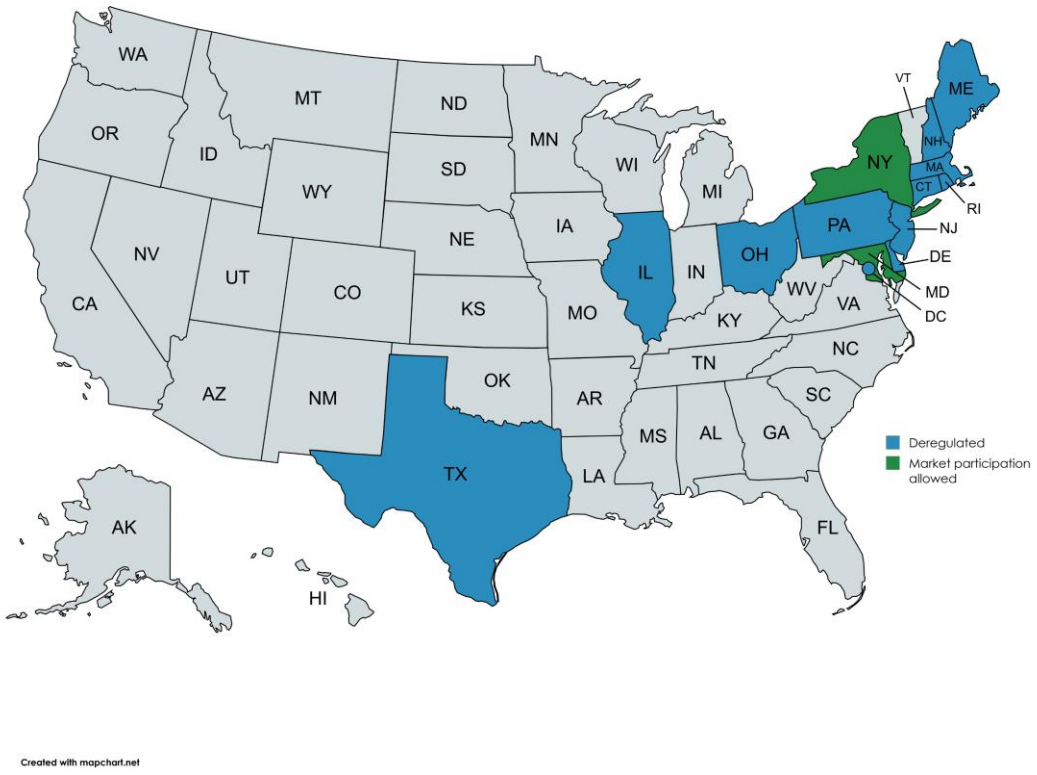
- 12 battery cycles for grid needs, 94 cycles for market participation.
- $23,121 from arbitrage and $71,114 from ancillary services, about **$8/kWh**.
- Battery offline for increased maintenance about 1 day per month.

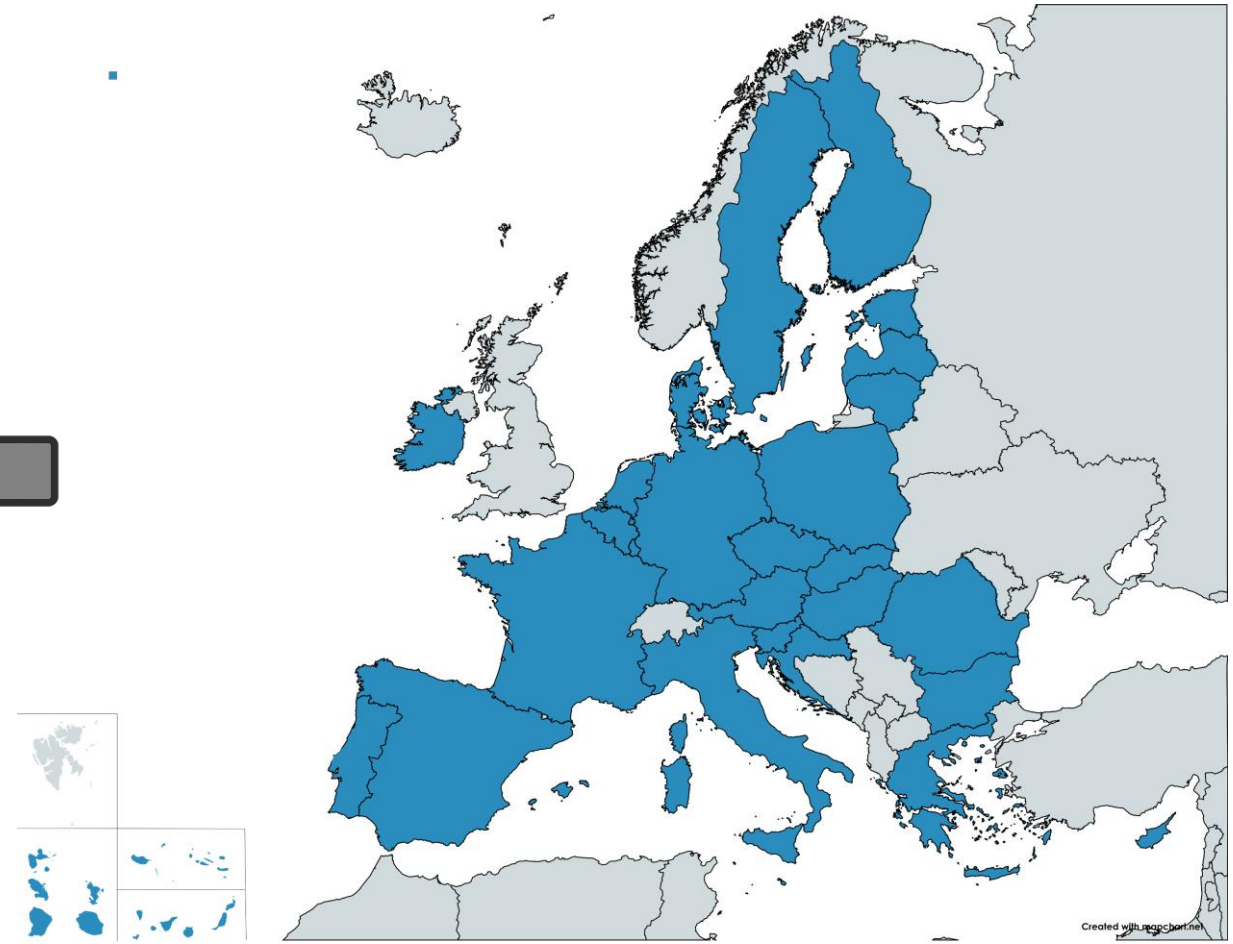
Geographical scope





Geographical scope





States allowing for market participation:

- MD (2019)
- NY (2021)

- EC (2023): Explore full flexibility potential of energy storage in distribution grids.
- ACER (2025): Create a market for local services that help alleviate distribution grid constraints.

Research program

Quantify the value of **investment deferral** (the grid value) and **arbitrage** and **scarcity supply** (the market value).

1. **Formal model** of market participation constraints for **battery operations** and **investment planning**.
2. **Compare** the grid value to the market value.
3. **Determine storage investments appropriate for distribution needs**.

Inform **state policy**.

Why arbitrage and capacity markets?

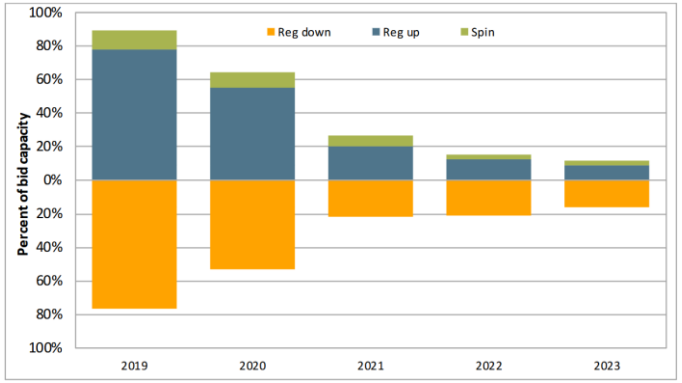


Figure 2.21: CAISO Market Monitor Special Report on Battery Storage, 2024

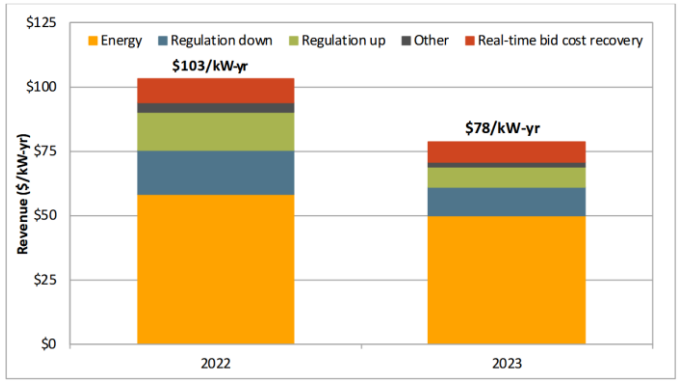


Figure 2.19: CAISO Market Monitor Special Report on Battery Storage, 2024

... because ancillary services markets are saturating (Figgener et al., 2023; Mastropietro et al., 2024).

State of the Art

Actual Operations (Orange & Rockland, 2024)

On a **daily basis**, decide if battery is needed for distribution grid support.

If not, decide if it should be used for market participation.

Academic Studies (PNNL, 2019)

During potential **N-1 contingency hours** maintain state-of-charge at 100%, discharge if contingency arises.

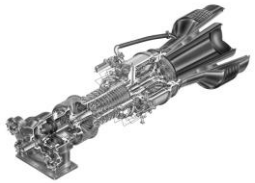
Maximum annual throughput of 150 cycles to maintain warranty listed as further study.

Our Contributions

Model both contingency and noncontingency scenarios. On a daily basis, decide on hourly operations for both scenarios. No need for state-of-charge heuristics. Include storage market participation in planning problems.

Problem Formulation (Operations / Economic Dispatch)

Supply Resources: $\mathcal{R} = \{\mathrm{b}, \mathrm{g}, \mathrm{s}\}$



backup

grid

storage

Demand Resources: $\mathcal{D} = \{\mathrm{g}, \ell, \mathrm{s}\}$

grid

load

storage

Production Decisions: $y^{\mathrm{s}}_{rk}\ \forall r \in \mathcal{R}, k \in \{1, \dots, K\}$

Demand Decisions: $y^{\mathrm{d}}_{rk}\ \forall r \in \mathcal{D}, k \in \{1, \dots, K\}$

Objective: Minimize linear operating costs.

Constraints (standard)**:** Capacity restrictions, nonnegativity, energy balance, charging and discharging losses.

Constraints (**Market participation**)**:**

$$\sum_{r \in \{\mathrm{b},\mathrm{s}\}} y^{\mathrm{s}}_{rk} \leq \left[y^{\mathrm{d}}_{\ell k} - \bar{x}_{\mathrm{g}} \right]^{+} \quad \text{(limited backup and storage)}$$

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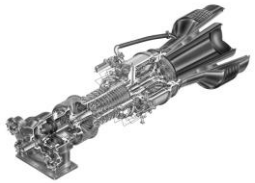

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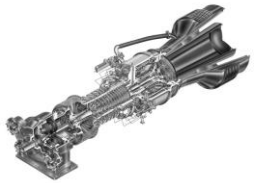
$$\sum_{r \in \{\mathrm{b},\mathrm{s}\}} y^{\mathrm{s}}_{rk} \leq \left[y^{\mathrm{d}}_{\ell k} - \bar{x}_{\mathrm{g}}\right]^{+} \text{ (limited backup and storage)} \iff \max\left\{y^{\mathrm{d}}_{\ell k} - \bar{x}_{\mathrm{g}}, 0\right\} + \sum_{r \in \{\mathrm{b},\mathrm{s}\}} y^{\mathrm{s}}_{rk} \geq 0$$

Problem Formulation (Investment decisions)

Supply Resources: $\mathcal{R} = \{\mathrm{b}, \mathrm{g}, \mathrm{s}\}$



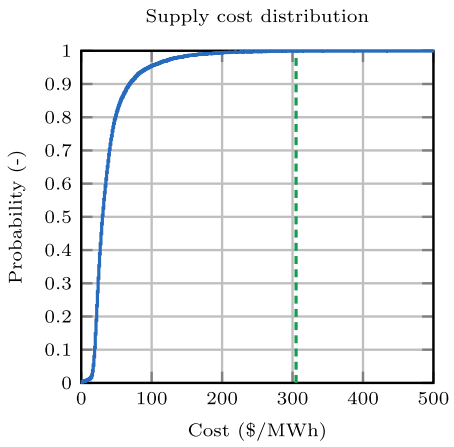
backup

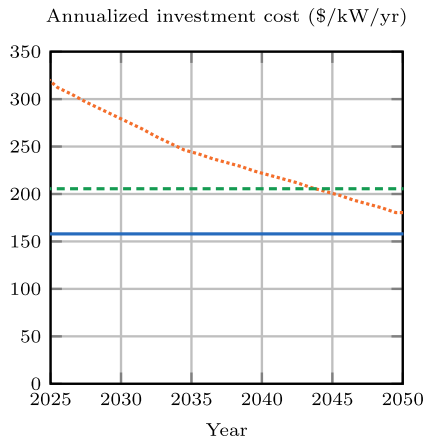
grid

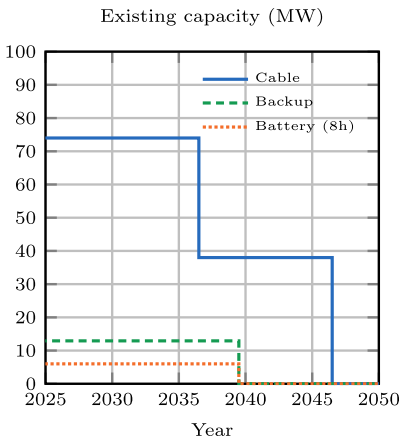
storage

Investment Decisions: $x_{rn}\ \forall r \in \mathcal{R}, n \in \{1, \dots, N\}$

Sample investment costs and existing capacity.







Objective: Minimize nonconvex investment costs.

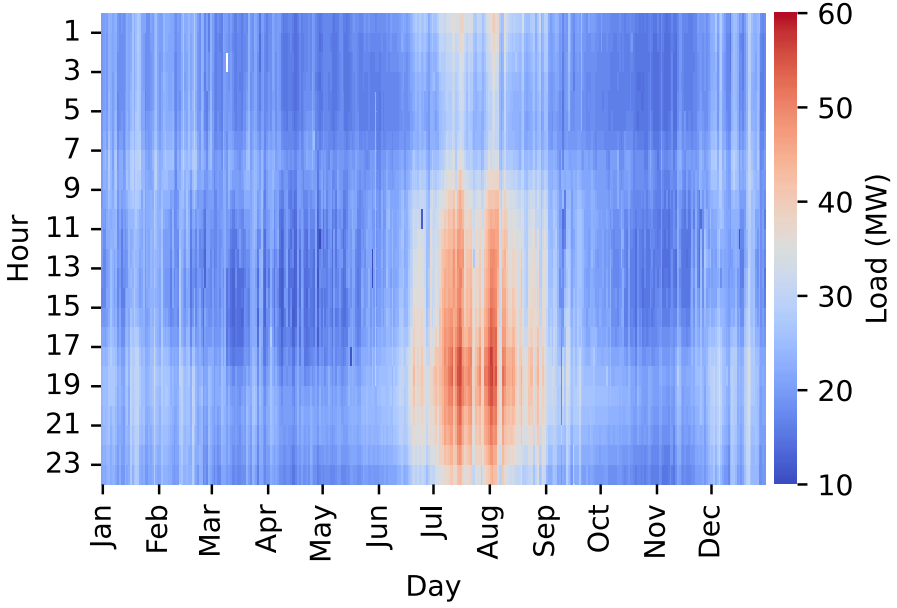
Constraints (standard)**:** Investment range, nonnegativity, lifetime, contingency-adjusted capacity.

Trade-off:

Lumpy investments with low per-unit costs vs modular investments with high-per unit costs.

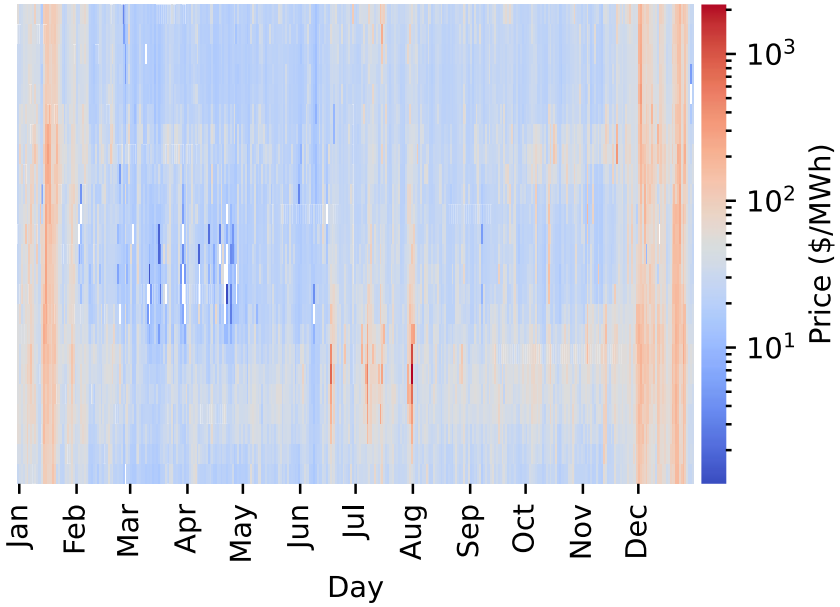
Case Study (Nantucket)

Electricity Demand (2024)



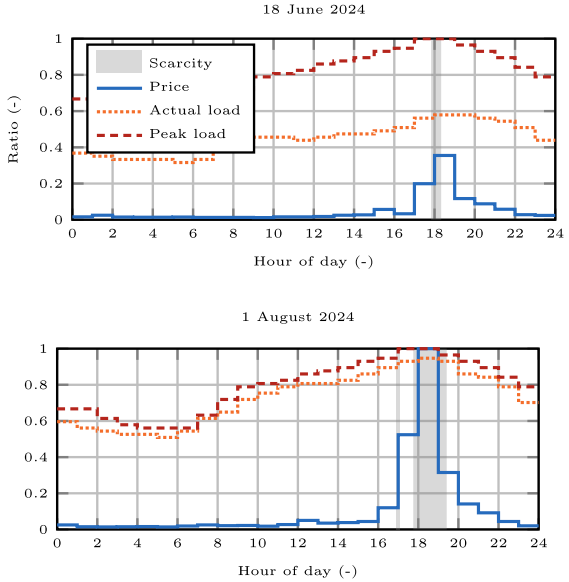
Demand peaks on summer afternoons.

Electricity Price (2024)



Price peaks on summer afternoons.

Scarcity Events (2024)



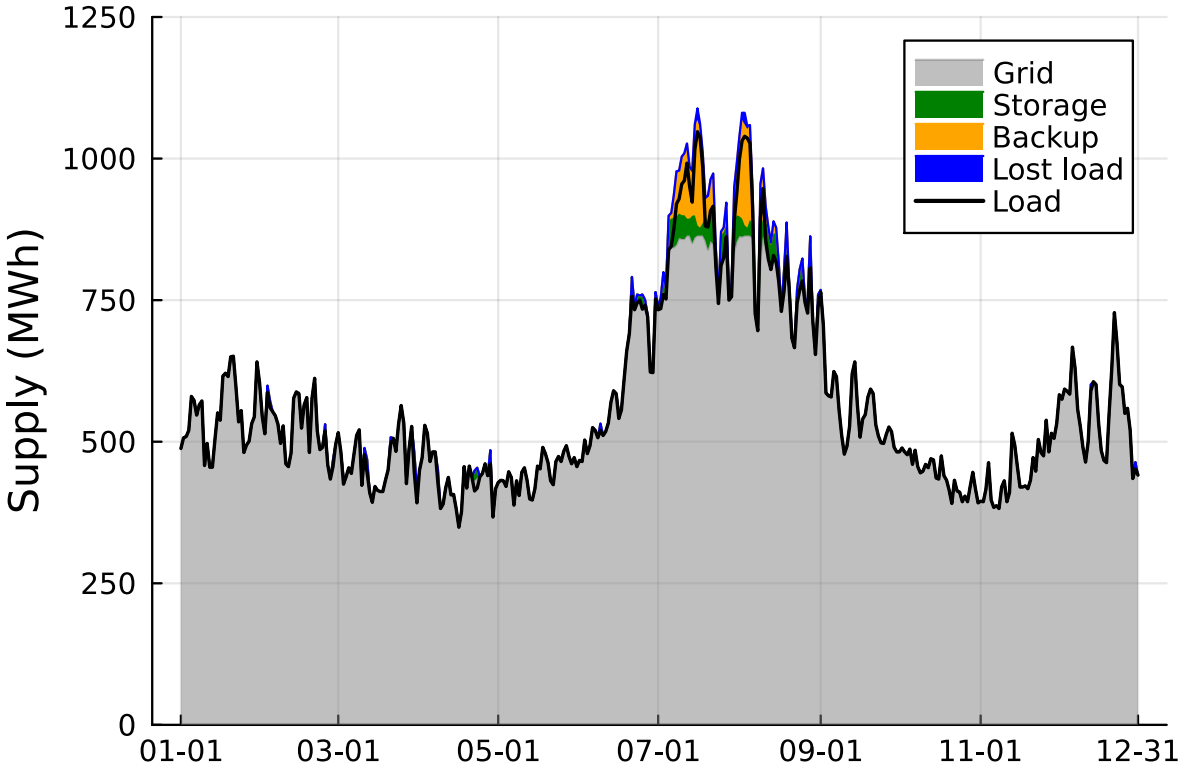
Scarcity happens on summer afternoons.

Analyze:

- Contingency supply mix with and without market participation.

Daily Supply Mix (Backtest 2024)

Limited Backup and Storage (Status Quo)

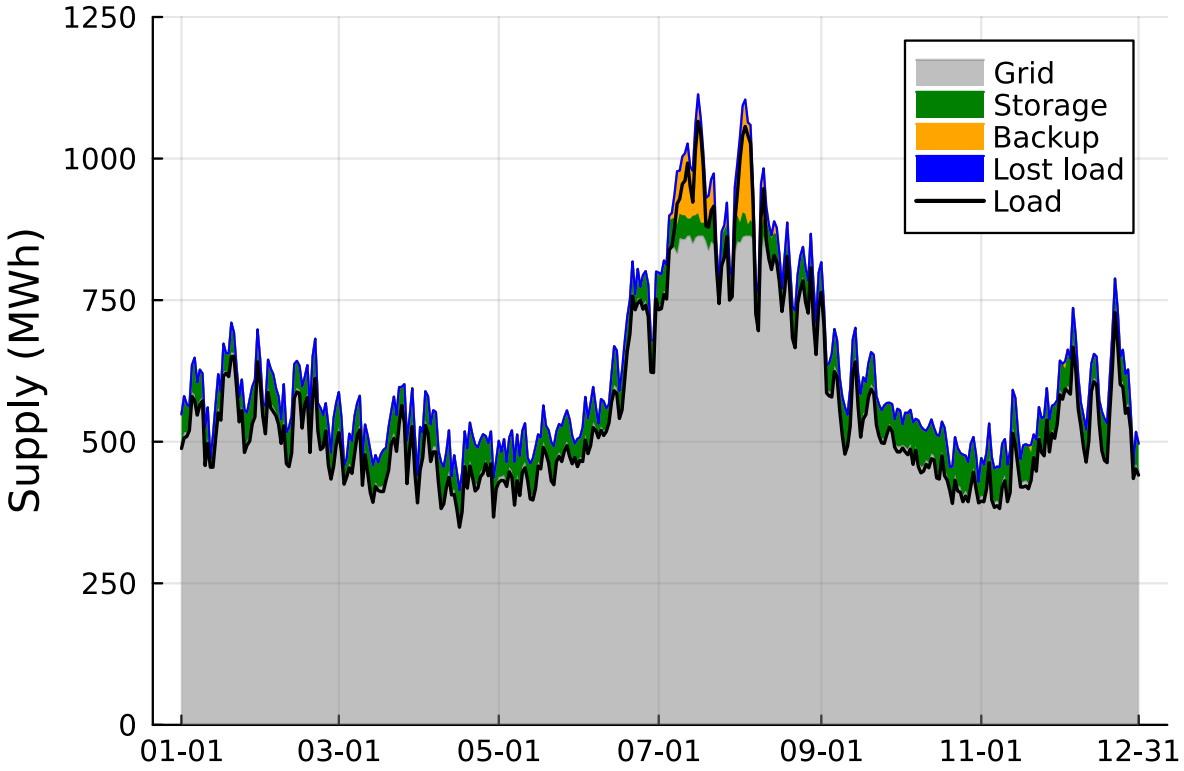


Storage is only used at times of peak demand.

Storage Supply: 1.7GWh (35 cycles)

Operating Cost: $M 9.6

Full Market Participation

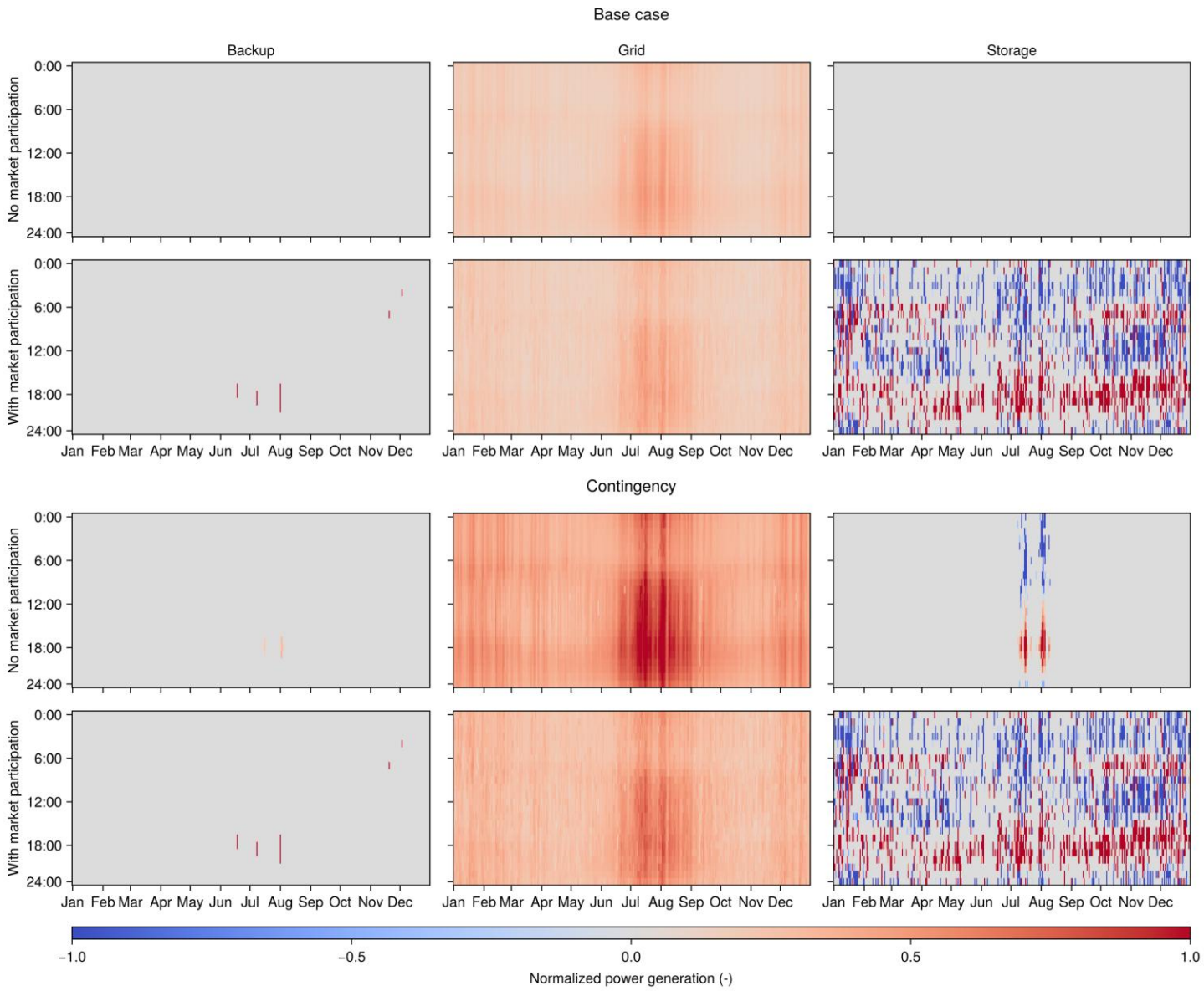


Storage used throughout the year.

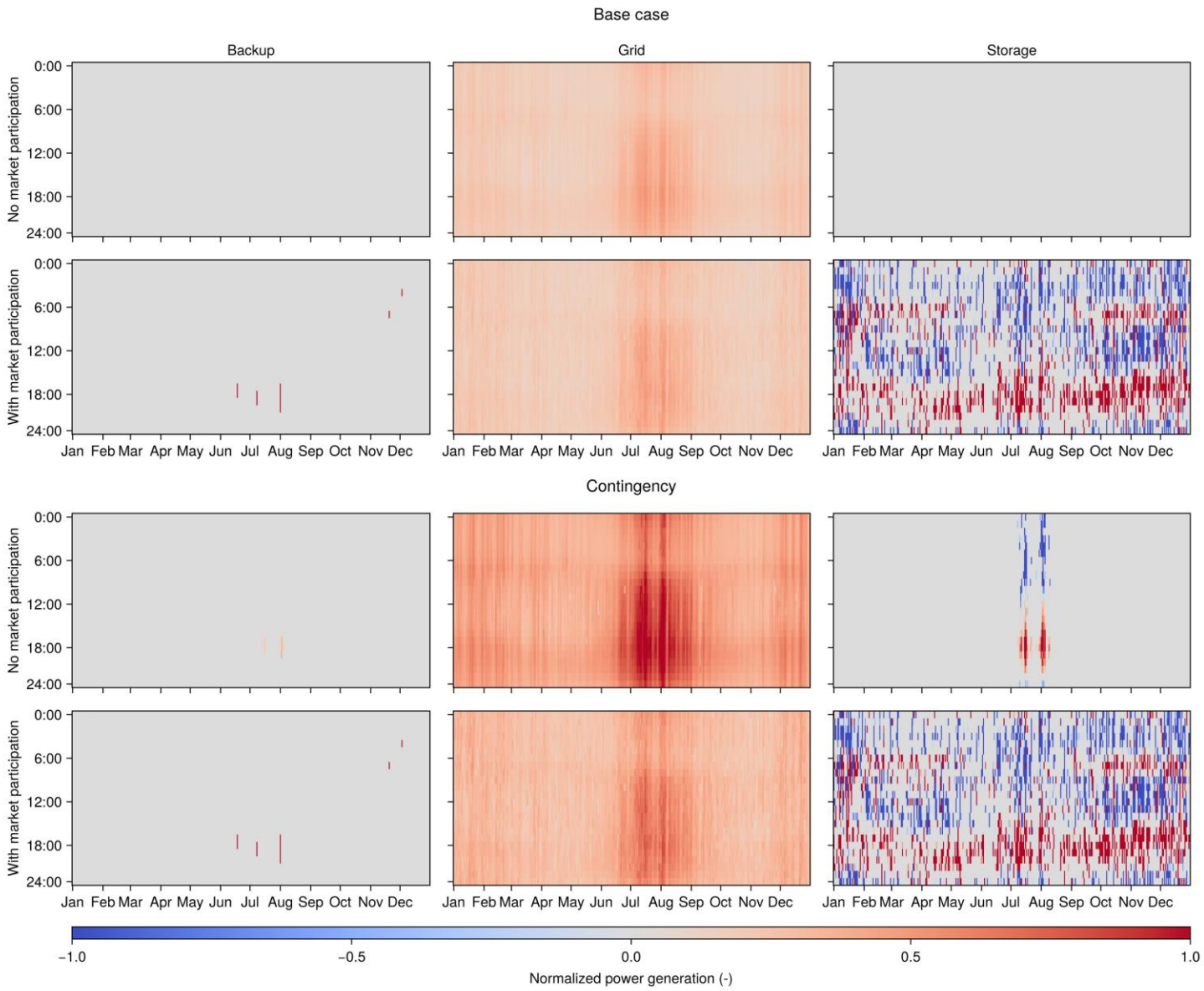
Storage Supply: 15.9GWh (331 cycles)

Operating Cost: $M 9.2

Resource Utilization (2025)

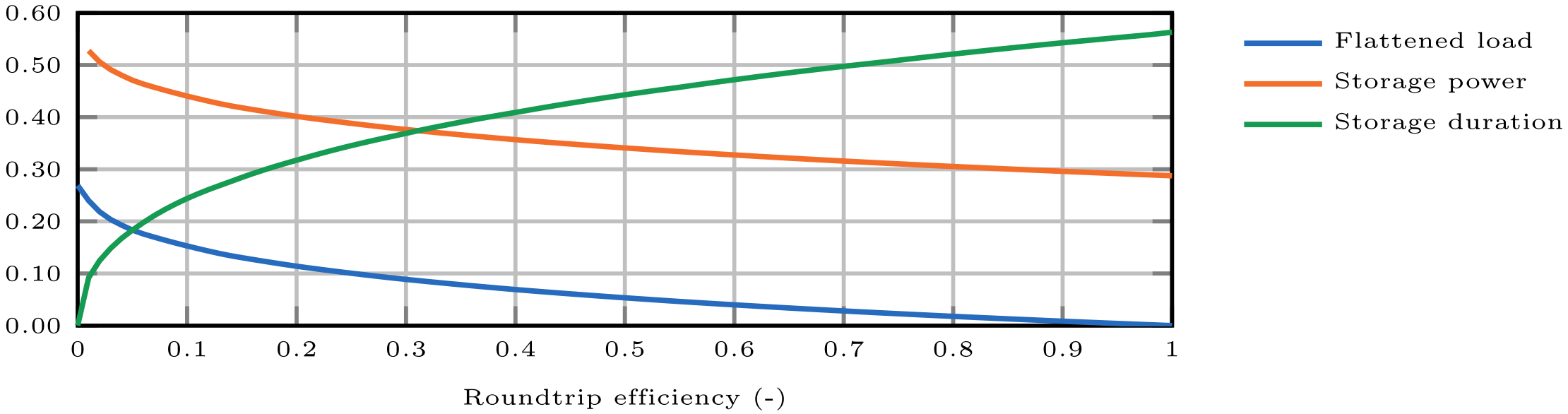


Resource Utilization (2025)



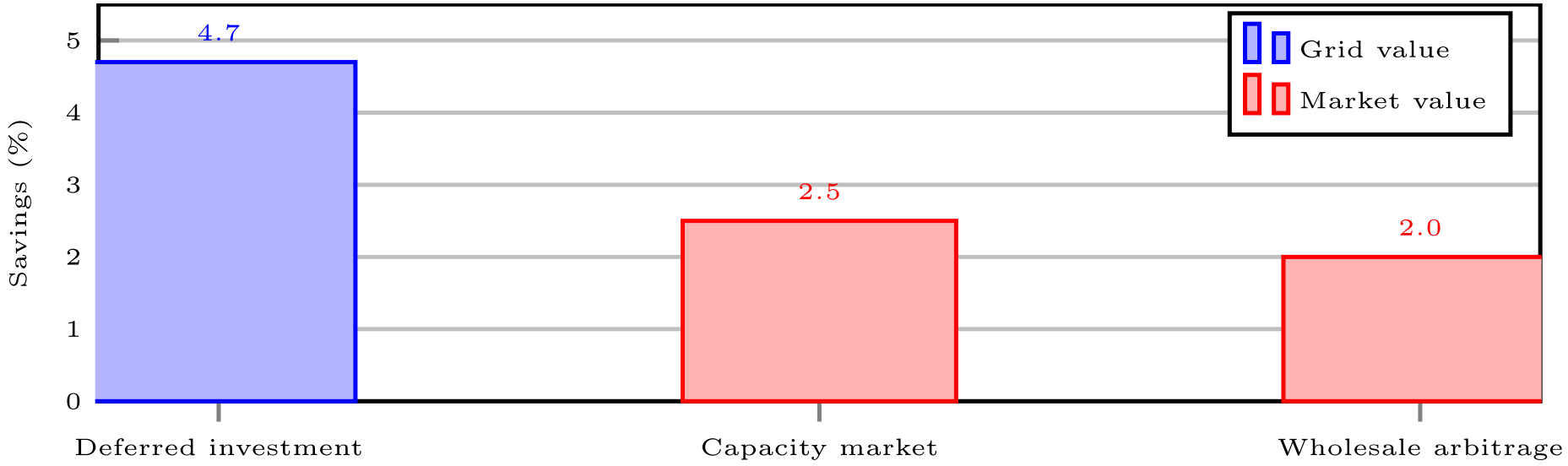
Peak Shaving Potential via Linear Programming.

Maximum peak shaving potential can be determined by a linear program.



Flattened load (normalized by the deviation from average load), storage power (normalized by average load), and storage duration (normalized by 12 hours).

Results and Conclusion



1. **Formal model** of market participation constraints for **battery operations** and **investment planning**. → Limit generation to cover shortfalls of grid capacity relative to peak demand (MILP).
2. **Compare** the grid value to the market value. → Similar.
3. **Determine storage investments appropriate for distribution needs**. → Via mathematical optimization (MILP or LP approximation). Under current market conditions, we find no incentive for extra investment.